CODI vs KaVa

1 BACKGROUND

Explicit chain of thought models write each reasoning step before producing an answer. This often improves accuracy, but it requires many generated tokens. Latent reasoning replaces those text tokens with a smaller number of continuous internal states.

The main difficulty is supervision. The answer alone gives little guidance about what each internal state should contain. CODI and KaVa provide that guidance in different ways.

CODI

CODI uses one model as both teacher and student. The teacher sees the written reasoning trace. The student solves the same problem using continuous states. Training matches the teacher and student hidden states near the answer position.

```
CODI loss = student answer loss + teacher loss +
hidden state loss
```

KaVa

KaVa adds supervision from the teacher key and value cache. The full teacher trace is compressed and matched to the student latent trajectory. This gives supervision at several internal positions rather than only at the answer boundary.

```
KaVa loss = CODI loss + key and value trajectory loss
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2 PURPOSE OF THE PILOT

The pilot was built to compare the two supervision styles under one shared implementation. Both methods started from the same pretrained GPT-2 revision. They used the same data, latent length, optimizer, schedule, prompts, decoding, and training budget. The main planned difference was whether the key and value loss was active.

This design was useful for testing the code and producing a matched comparison. It was not a faithful reproduction of the original KaVa architecture.

3 PILOT SETUP AND TRAINING PROCEDURE

Component	Setting
Backbone	Pretrained GPT-2
Training data	385,620 GSM8K-Aug equation examples
Latent states	Six autoregressive states
Training	One epoch and 96,405 optimizer steps
Batch size	Four
Learning rate	1e-4 with cosine decay
Evaluation	Greedy decoding and numeric exact match

How each training step worked

1. A deterministic seeded batcher selected four equation trace examples. Inputs were limited to 256 tokens.
2. The same GPT-2 parameters served as teacher and student. The teacher received the question, written equation trace, and answer. The student received the question and produced six continuous latent states before predicting the answer.
3. Both methods optimized student answer loss, teacher language model loss, and L1 hidden state matching across all transformer layers at the answer endpoint.
4. KaVa additionally matched compressed teacher keys and values to the six student latent positions across all layers. CODI used the identical training path with the KV loss weight set to zero.
5. AdamW updated all model parameters. The learning rate warmed up for 500 steps to 1e-4 and then followed cosine decay. Weight decay was 0.1 and gradient norm was clipped at 2.0.

Saving and resuming

A checkpoint was written every 1,000 steps and the last two were retained. Each checkpoint stored model weights, optimizer state, random state, loss history, and the experiment fingerprint. A nine hour wall clock guard saved the current state before notebook limits were reached. Kaggle and Colab sessions resumed from the latest verified checkpoint until step 96,405.

Evaluation data

Dataset	Full size	Role
GSM8K	1,319	In domain arithmetic
GSM-Hard	1,319	Harder numbers
SVAMP	300	Word problem transfer
MultiArith	180	Multi step arithmetic

Runs completed

- CODI and KaVa at seeds zero, one, and two
- No trajectory distillation at seed zero
- Random and uniform compression at seed zero
- Zero, mean, and cross example latent interventions
- A position by position KaVa shuffle sweep

Metric and uncertainty

Accuracy is numeric exact match. Macro accuracy is the plain average across the four datasets. The primary comparison used 10,000 paired bootstrap samples. This measures uncertainty from the test questions for the fixed checkpoints. It does not test whether the checkpoints were trained well enough.

4 KAVA SCORED HIGHER IN THE PILOT

KaVa scored higher on GSM8K, MultiArith, and SVAMP. CODI was slightly higher on GSM-Hard. The largest difference was on MultiArith.

Figure 1 Full seed zero accuracy

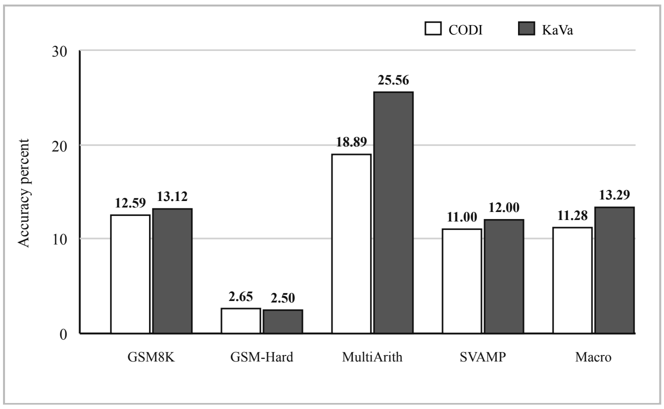

Figure 1 Numeric exact match on the complete seed zero evaluation. KaVa gained 2.01 macro points. The 95 percent paired interval was 0.44 to 3.60 points.

Task level result

Dataset	CODI	KaVa	Gain
GSM8K	12.59	13.12	0.53
GSM-Hard	2.65	2.50	-0.15
MultiArith	18.89	25.56	6.67
SVAMP	11.00	12.00	1.00

Only the MultiArith interval excluded zero. This dataset produced most of the macro gain.

5 THE DIRECTION HELD ACROSS SEEDS

Seed	CODI	KaVa	Gain
Zero	11.72	13.89	2.17
One	11.78	12.75	0.97
Two	11.53	12.61	1.08
Mean	11.68	13.08	1.41

KaVa scored higher at each seed. Three seeds show a stable direction, but they do not provide a precise method level confidence interval.

6 CONTROLS DID NOT ISOLATE R-KV

Seed zero run	Macro
CODI	11.72
No trajectory loss	13.19
Uniform compression	12.69
Random compression	13.40
R-KV KaVa	13.89

Full KaVa scored highest. Its differences from the other controls were uncertain. The pilot therefore did not prove that R-KV selection caused the gain.

7 LATENT DAMAGE AFFECTED KAVA MORE

The same latent damage was applied to both models. Negative values below mean that KaVa lost more accuracy than CODI.

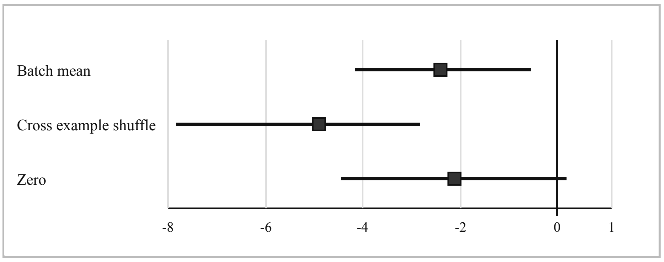

Figure 2 Difference in differences with 95 percent paired intervals. Cross example shuffling harmed KaVa by 4.60 points more than CODI.

> **Reliability limit**
> These interventions show how the pilot checkpoints behave. They do not show how accurate paper level checkpoints would behave.

8 PUBLISHED CODI IS MUCH STRONGER

The CODI paper trained GPT-2 on GSM8K-Aug for 40 epochs. It used an effective batch size of 128, a learning rate of 3e-3, LoRA rank 128, and bfloat16. A single GPT-2 run took about 36 hours on one A100 with 80 GB memory.

Model	GSM8K	Hard	SVAMP	MultiArith
Pilot CODI	12.59	2.65	11.00	18.89
Paper GPT-2	43.7	9.9	42.9	92.8
Paper LLaMA 1B	55.6	12.8	61.1	96.1

The paper GPT-2 model was 31.11 points higher on GSM8K and 73.91 points higher on MultiArith. This gap is too large to treat the pilot as a reproduction.

Figure 3 Pilot CODI compared with paper CODI

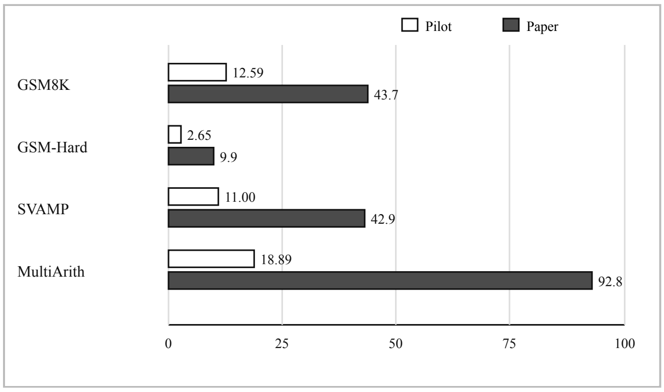

Figure 3 The pilot CODI checkpoint is far below the published GPT-2 checkpoint on every shared dataset.

9 PUBLISHED KAVA USES A DIFFERENT SETTING

The KaVa paper did not evaluate GPT-2. It used Qwen 0.5B and LLaMA 1B and 3B instruction models. Training used LoRA, 24 parallel latent states, and three Jacobi iterations. The 0.5B and 1B models were trained for ten epochs with effective batch size 128.

Backbone	Method	GSM8K	Hard	SVAMP
Qwen 0.5B	CODI	37.5	8.1	47.0
Qwen 0.5B	KaVa	46.9	10.8	50.6
LLaMA 1B	CODI	53.9	12.6	59.0
LLaMA 1B	KaVa	56.5	12.7	58.9
LLaMA 3B	CODI	61.0	15.0	72.4
LLaMA 3B	KaVa	65.7	15.2	72.7

The strongest KaVa gain appeared when the training traces used natural language. With Qwen 0.5B, CODI scored 20.2 on GSM8K and KaVa scored 44.4. This result provides a narrow direction for the next stage.

10 WHY THE PILOT WAS UNDERTRAINED

Factor	Pilot	Published setting
Data exposure	One epoch	Forty CODI or ten KaVa epochs
Effective batch	Four	128
Backbone	GPT-2	GPT-2 CODI or 0.5B to 3B KaVa
KaVa states	Six autoregressive	Twenty four parallel states
KaVa updates	Autoregressive	Three Jacobi iterations

Compute and time constraints

The pilot was designed around limited Kaggle and Colab sessions. GPU access was not continuous. Each full checkpoint was about 1.4 GB, so saving, moving, and resuming many runs also took time. A paper level CODI GPT-2 run takes about 36 hours on an A100. The published KaVa setting needs larger models, ten passes through 385,620 examples, and several teacher and student forward passes per batch.

These limits led to the one epoch design and the smaller backbone. That choice was reasonable for code validation and a first matched experiment. It was not enough for a final accuracy comparison. The report should have made this distinction earlier.

Checkpoint availability

Official CODI weights are available for GPT-2 and LLaMA 1B. They should have been used as the first evaluation check. No official KaVa checkpoint or complete training repository was found in the paper resources.